

From Pitcher Fatigue to Market Efficiency: An Empirical Evaluation of Public Information in Live MLB Totals Markets

This paper asks whether publicly observable baseball information predicts outcomes, or merely predicts what the market already knows.

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Abstract

Live, in-play wagering is a large and growing share of sports-betting volume, yet market-efficiency research remains concentrated on pre-game prices; the calibration of live markets, and their incorporation of in-game information, are comparatively unstudied. We ask a single question: do publicly observable baseball state variables contain incremental predictive information about remaining runs beyond the forecast embedded in a sharp live betting market? Using 163 Major League Baseball games (June 2026) with one-minute live-total trajectories from a sharp book, pitch-level measurement, and full play-by-play, we subject a battery of public hypotheses — times through the order, velocity decline, bullpen fatigue, line-drop reversion, alternate-line skew, early-run anchoring, and weather/park effects — to an escalating validation protocol ending in a forecast-encompassing test against the market itself. No variable survives. The market's forecast error is not predictable from any variable we measure (out-of-sample $R^2 = -0.037$); an apparent velocity signal is shown to be selection bias; and an event-level transfer function indicates the line adjusts to information shocks by an approximately uniform, near-correct magnitude. Within our data we find no evidence of exploitable public-information inefficiency, and we characterize the boundary precisely. We release a transferable validation protocol and a reproducible benchmark.

Keywords: market efficiency; in-play betting; forecast encompassing; calibration; incremental information; reproducible benchmark. **JEL codes:** C53 (forecasting), G14 (information and market efficiency), Z23 (sports economics).

1. Introduction

Live, in-play wagering has grown from a marginal product into a large and rising share of sportsbook handle — industry reporting consistently places it at or near half of volume in mature markets. Yet the empirical study of wagering-market efficiency, a literature surveyed by Sauer (1998), has concentrated on pre-game moneylines and point spreads, where a single closing price can be compared against a realized outcome. Live markets pose a different and less-studied problem: prices update continuously as the game state evolves, so the object of interest is not a single forecast but a stream of them, and the natural questions concern calibration and information

incorporation rather than a one-shot verdict. Whether live markets efficiently absorb the publicly observable information generated during a game — and how one would even test such a claim — remains comparatively open.

Baseball is an unusually favorable setting in which to ask the question. The game unfolds as a sequence of discrete events with well-established run values (the RE24 base-out run-expectancy table and linear weights), so the informational content of each event can be quantified rather than estimated. Pitch-level measurement makes within-game state — velocity, pitch count, times through the order — observable in fine detail. And the sport carries a strong body of public prior belief, most prominently the times-through-order penalty (Tango, Lichtman, and Dolphin, 2007): the widely held view that a starting pitcher degrades sharply the third time he faces a lineup, which, if under-priced, would make the live Over a profitable bet. That belief is a natural entry point — a concrete, popular, plausible hypothesis with which to probe the general question.

The general question, however, is not whether such variables predict scoring. Many of them do. It is whether they carry *incremental* predictive information once the forecast already embedded in a sharp live market is conditioned upon. This distinction — between predicting an outcome and adding information beyond an existing forecast — is the conceptual center of the paper, and it is routinely elided in applied betting research, where an in-sample association is often treated as evidence of an edge. We take the opposite stance: an in-sample signal is the weakest possible evidence, and each candidate variable must survive progressively stronger tests, culminating in a forecast-encompassing test against the market itself. Within the limits of our data, no variable does. The result is not a failure to find an edge but a characterization of the boundary at which publicly observable baseball information ceases to add value beyond the market — a boundary we map precisely, and bind to the conditions under which it was measured: 163 games across a single month, at one-minute cadence, from a single sharp-book feed.

Contributions

This paper makes three contributions.

1. Empirical. For a battery of publicly observable baseball variables — times through the order, velocity decline, bullpen fatigue, line-drop reversion, alternate-line skew, early-run anchoring, weather, and park — we show that none provides incremental predictive information about remaining runs once conditioned on the live market forecast, and we locate the point at which that incremental value disappears.

2. Methodological. We formalize an escalating validation protocol — signal, robustness, out-of-sample, debiasing, conditional testing, forecast encompassing, transfer function — that shifts the burden of proof from demonstrating *prediction* to demonstrating *incremental information beyond an existing forecast*. The protocol is domain-general and applies to any market with a sharp public forecast and observable state.

3. Infrastructure. We release the cleaned datasets, the feature schema, the evaluation protocol, and reference implementations of the market forecast, remaining-runs model, encompassing tests, and transfer function as the initial release of the **Third Turn Benchmark**, so that future hypotheses can be evaluated against the same reference rather than re-derived from scratch.

The remainder of the paper is organized around a single question, which every subsequent section serves:

Do publicly observable baseball state variables contain incremental predictive information about remaining runs beyond the forecast embedded in a sharp live betting market?

2. Related Work

This section is deliberately brief: its job is to establish a gap, not to survey three literatures. We draw on each only for what the design requires.

Wagering-market efficiency. The economics of wagering markets is a mature literature, surveyed by Sauer (1998), whose broad finding is that betting prices are accurate but not perfect: systematic deviations such as the favorite–longshot bias exist, yet rarely survive transaction costs. For baseball specifically, Woodland and Woodland (1994) document a *reverse* favorite–longshot bias in the moneyline market and conclude that the deviations are insufficient for profitable betting. This literature is overwhelmingly pre-game. The in-play evidence is thinner and mixed. Croxson and Reade (2014) find that soccer exchange prices update swiftly and essentially fully at goal arrival, consistent with semi-strong efficiency. More recent work documents imperfections in the price *process* itself: Simon (2024) rejects weak-form efficiency for MLB moneyline movement, finding systematic overreaction; Simon (2025) generalizes the negative autocorrelation of price changes to NFL, NBA, and NHL markets; and Angelini and De Angelis (2026) measure an approximately 0.64-for-one contemporaneous underreaction to benchmark-probability changes in real-time prediction markets, with predictable subsequent drift.

Baseball performance. The times-through-order penalty entered the sabermetric canon through Tango, Lichtman, and Dolphin (2007). The strongest recent evidence, however, favors continuous decay over a discontinuity: Brill, Deshpande, and Wyner (2023) find little support for a performance cliff at the third time through the order once batter and pitcher quality and other confounders are controlled — a conclusion our own gradient analysis independently reproduces. Velocity effects on batting outcomes are real but small per mile per hour: Sutton-Brown (2023) estimates roughly 0.004 wOBA per mph operating through pitch quality, with a residual direct effect an order of magnitude smaller. Run values for discrete events (the RE24 base-out run-expectancy table and linear weights) are standard tools from the same tradition.

Forecast evaluation. Our central test imports the forecast-encompassing framework of Chong and Hendry (1986) from macroeconomic forecast evaluation: a benchmark forecast encompasses a rival when the rival adds no explanatory power for the target beyond the benchmark. Calibration diagnostics — reliability curves, the Brier score, expected calibration error — and the probabilistic interpretation of the area under the ROC curve (Hanley and McNeil, 1982) complete the evaluation toolkit.

The gap. These strands study either the price series in isolation, a single discrete shock, or baseball performance without reference to prices. To our knowledge, no published work combines live baseball state, pitch-level features, calibration analysis, an event-level transfer function, and a forecast-encompassing test in which a sharp live betting line serves as the benchmark forecast. That combination is this paper's contribution.

3. Methods

Our design compares two forecasts of the same quantity — the number of runs a game has left to score at a given moment — and asks whether publicly observable state improves the forecast already implied by the market. The unit of analysis throughout is the *half-inning snapshot*: a single moment in a single game at which both the market's live total and the full game state are observed. This section defines the data behind each snapshot (§3.1), how state variables are constructed from it (§3.2), the escalating protocol by which a candidate variable is tested (§3.3), the statistical estimators applied at each rung (§3.4, including why forecast encompassing is the central test), and the artifacts released for reproduction (§3.5).

3.1 Data

The study draws on 163 Major League Baseball games played in June 2026, each observed from three aligned sources. **Market prices.** Live full-game total (Over/Under) lines and their associated prices were recorded as one-minute trajectories from a single Pinnacle-grade feed, giving, for each game, the market's evolving point forecast of final total runs and its implied Over probability. **Game state.** Complete play-by-play and boxscore records from the MLB Stats API supply, at every plate appearance, the inning and half, base-out state, score, batting-order position, the identity and pitch count of the pitcher, and the times each batter has faced the current starter. **Pitch measurement.** Pitch-level release speeds (`startSpeed`) from the same feed provide within-game velocity trajectories. Venue and weather (temperature, wind speed and direction relative to the field) and each game's realized final total complete the record. Sources are joined on game identifier and, for odds, on timestamp. The one-minute cadence and single odds source are the principal constraints on what the design can measure; their consequences are stated in §6.

3.2 Feature construction

From the raw record we construct, at each snapshot, the two forecasts under comparison and the public state variables that might improve on them. The **market forecast of remaining runs** is $B = (\text{live total}) - (\text{runs already scored})$; the **realized remaining runs** is $Y = (\text{final total}) - (\text{runs already scored})$. $Y - B$ is therefore the market's forecast error at that snapshot — the object whose predictability is the crux of the study.

The candidate state variables are built without reference to the outcome. The starter is the pitcher at the first plate appearance of each half; batting-order slots are assigned by order of first appearance against that starter, and times-through-order counts prior faced batters divided by nine. Velocity decline is computed two ways — across successive times through the order, and within a fixed early pitch-count window (first twenty pitches versus the next twenty) — a distinction that becomes decisive in §3.3. Bullpen quality is each team's season relief runs allowed per nine innings; park factor and signed wind/temperature come from static published tables. The true change in run expectancy at each event, used by the transfer function, is $\Delta RE = (\text{runs scored}) + (RE24_{\text{after}} - RE24_{\text{before}})$, where RE24 is the standard 24-state base-out run-expectancy table (Tango, Lichtman, and Dolphin, 2007). No feature uses information unavailable at the snapshot it describes.

3.3 Validation protocol

The core of the design is a fixed sequence of tests of *increasing stringency*, applied in the same order to every candidate variable. A variable is carried forward only until it is eliminated, and we report the rung at which elimination occurs; a variable that reaches the top has been tested against progressively harder alternatives rather than a single easy one. The ladder is:

Signal → Robustness → Out-of-sample → Debiasing → Conditional testing → Forecast encompassing → Transfer function

Signal asks whether the variable is associated with scoring at all, in-sample. **Robustness** asks whether that association survives reasonable changes in specification, thresholds, and sub-sample — an in-sample edge that moves with an arbitrary cutoff is rejected here. **Out-of-sample** re-estimates every model by leave-one-game-out cross-validation, so a variable is judged only on games not used to fit it. **Debiasing** replaces any post-treatment measurement — one defined only on a subsample selected by the outcome — with a pre-treatment analogue; a signal that survives in-sample but vanishes once measured before treatment is diagnosed as selection, not effect. **Conditional testing** asks whether the variable earns its keep only within specific contexts (e.g. hitter-friendly parks or weather) rather than on average. **Forecast encompassing** conditions on the market forecast itself and asks whether the variable adds anything to it. **Transfer function** finally asks not whether the market prices the variable but whether it prices it by the correct magnitude. The protocol embodies a single guiding principle — *a betting hypothesis should be evaluated against the market, not merely against the outcome* — and the last two rungs are what enforce it. Figure 1 traces the study's actual course through this sequence.

The research process: each surviving explanation handed to a stricter test

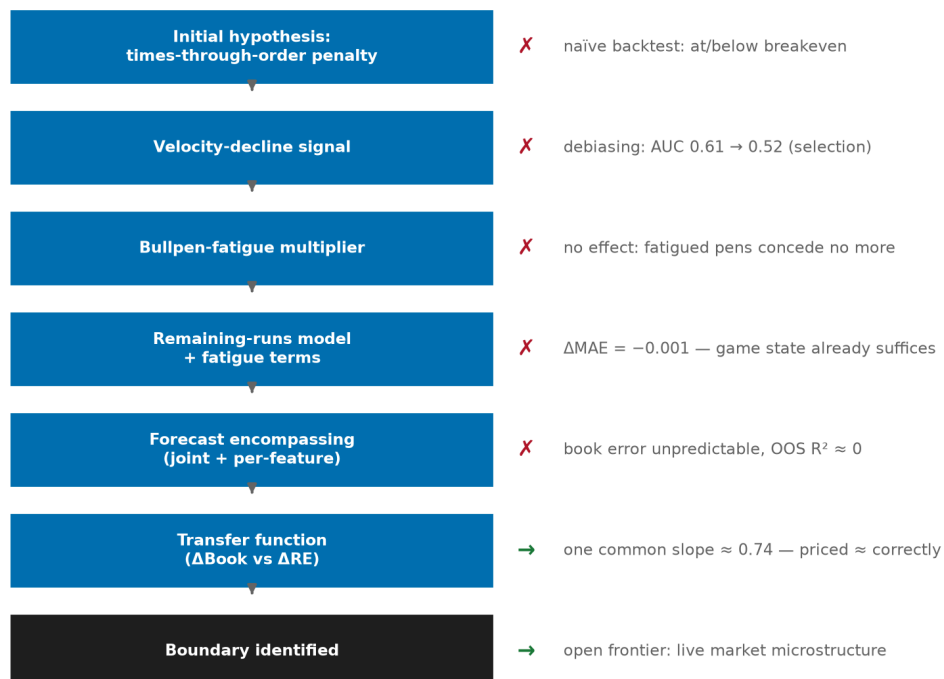


Figure 1. The research process. Each surviving explanation was handed to a stricter test; the sequence terminates at a boundary, not at an edge.

3.4 Statistical evaluation

Each rung is a specific estimator, all computed out-of-sample by leave-one-game-out.

Forecast encompassing (Chong and Hendry, 1986). For remaining runs Y , market forecast B , and a candidate feature set X , we fit three ridge-regularized linear forecasts — $Y \sim B$, $Y \sim X$, and $Y \sim B + X$ — and compare their out-of-sample R^2 and mean absolute error. If $Y \sim B + X$ does not improve on $Y \sim B$, the market encompasses X . The sharpest form regresses the market's forecast error $Y - B$ directly on X : if X cannot predict the error out-of-sample, it carries no information the price lacks. A per-feature variant (E+) fits $Y \sim B + X_i$ against $Y \sim B$ for each variable individually, so that two proxies for the same state cannot mask one another in the joint model. Continuous predictors are standardized; the ridge penalty is fixed a priori and applied to all non-intercept terms.

Why forecast encompassing? Ordinary predictive accuracy cannot separate the two claims this paper must distinguish. A model that predicts runs well demonstrates only that a variable is informative about the outcome; it says nothing about whether that information is already contained in an existing forecast. Forecast encompassing answers the second question directly by conditioning on the market forecast before evaluating the variable, so that the quantity being tested is *incremental* information rather than raw predictive power. Because a variable can predict the outcome while adding nothing beyond the market — the empirical situation we in fact observe — this conditioning is not a refinement of the analysis but its center.

Calibration. We assess the market forecast by binning snapshots on B and comparing mean realized Y per bin against the identity line, and we assess probabilistic forecasts (e.g. the probability a team exceeds a run threshold) with reliability curves, the Brier score, expected calibration error, and area under the ROC curve. The velocity debiasing of §3.3 is evaluated here as the change in out-of-sample AUC between a baseline forecast and forecasts augmented with the post-treatment versus pre-treatment velocity measures; AUC confidence intervals use the analytic variance of Hanley and McNeil (1982).

Transfer function. For each in-game event we pair the true change in run expectancy ΔRE with the converged change in the live total one and five minutes later ($\Delta Book$), and estimate the response ratio $\Delta Book / \Delta RE$ by event type together with a single common slope through the origin. Mean ΔRE by event type is checked against published linear weights as a validity control.

Uncertainty. All point forecasts are out-of-sample (leave-one-game-out); interval estimates use the Hanley–McNeil formula for AUC, Wilson intervals for proportions, and the bootstrap otherwise. Differences smaller than the width of their intervals are reported as such and are not interpreted as effects.

3.5 Reproducibility

Every quantity in this paper is recomputed from committed inputs by a fixed set of scripts; estimation is deterministic (leave-one-game-out folds and fixed seeds), so results regenerate exactly. To facilitate reproduction

we release the cleaned datasets, the feature schema, the evaluation protocol of §3.3, and reference implementations of the market forecast, the remaining-runs model, the encompassing tests, and the transfer function as the initial release of the **Third Turn Benchmark**, archived under a persistent DOI alongside the code repository and the research log. The frozen result artifacts (output/*.json) and the scripts that produce them are sufficient to reconstruct every figure and number without access to the original feeds.

4. Results

The research question is answered directly by Figure 4; the remaining figures explain why the answer takes the form it does.

Figure 4 demonstrates that publicly observable baseball variables provide no measurable incremental predictive information beyond the live market. Across 2,505 half-inning snapshots, a leave-one-game-out forecast of remaining runs from our full feature set achieves an out-of-sample R^2 of 0.279; the market-implied remaining total alone achieves 0.304; and combining the two changes R^2 by -0.017 — adding the features to the market does not improve, and slightly degrades, the forecast. The right panel shows the per-feature version of the same test (E+): each variable's individual incremental R^2 beyond the market falls inside a ± 0.003 band (best case, bullpen, at $+0.0018$; velocity, starter tier, temperature, wind, and park at or below zero), so no single variable hides behind the others. As the sharpest test, we regress the book's forecast error — realized minus market-implied remaining runs — directly on the features: it is not predictable out-of-sample, $R^2 = -0.037$. This is the central empirical result of the study.

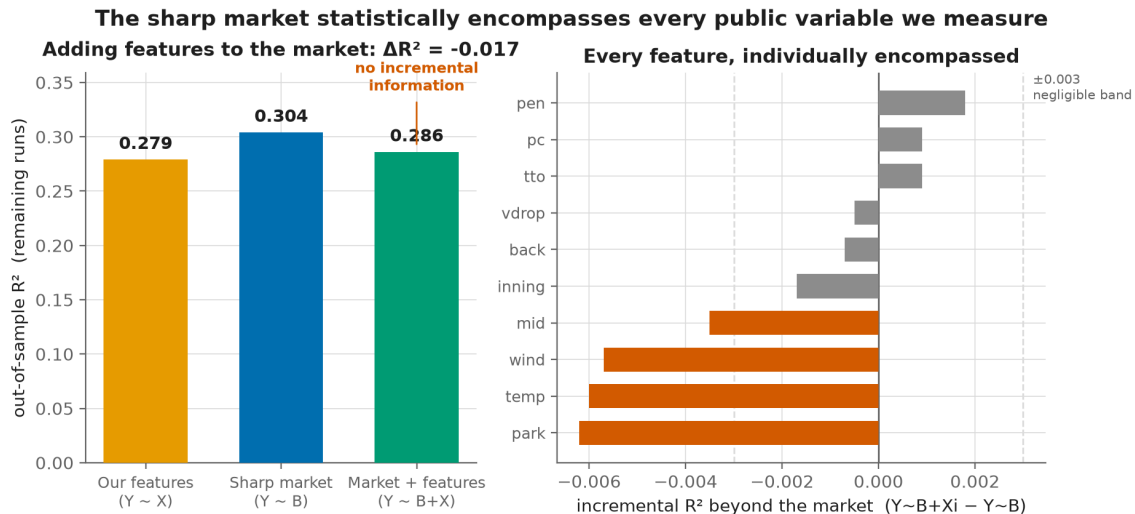


Figure 4. Forecast encompassing. *Left:* adding the feature set to the market forecast changes out-of-sample R^2 by -0.017 . *Right:* each feature's individual incremental R^2 beyond the market; bars within the ± 0.003 band are drawn neutral.

Figure 2 shows that this boundary is not a property of one variable but of the entire battery. The figure arranges ten candidate hypotheses drawn from the public handicapping literature — times-through-order, velocity decline, bullpen-fatigue multipliers, drop reversion on both sides, alternate-line skew, early-run anchoring, weather/park context, a remaining-runs fatigue term, and forecast encompassing — against five escalating gates: initial signal, robustness, out-of-sample, market test, verdict. The pattern of elimination is the finding: several hypotheses clear an in-sample signal, fewer survive robustness, fewer survive out-of-sample cross-validation, and none clears the market test. Because the gate of elimination varies by row — times-through-order and the velocity signal die at robustness, drop-reversion (Over) and forecast encompassing at out-of-sample, the context hypotheses at the market test — no single artifact (a coding error, one anomalous month, one mis-specified model) is a plausible common cause. The complete battery, with each hypothesis's motivation and mode of elimination, is summarized in Appendix Table A1.

Sequential elimination of candidate public-information hypotheses

	Initial signal	Robustness	Out-of- sample	Market test	Verdict
Times-through-order (TTOP)	✓	✗	▪	▪	✗
Velocity decline	✓	✗	▪	▪	✗
Bullpen-fatigue multiplier	✗	▪	▪	▪	✗
Drop reversion (Over)	✓	✓	✗	▪	✗
Drop reversion (Under)	✓	✗	▪	▪	✗
Alternate-line skew	✓	✓	✓	✗	✗
Early-run anchoring	✓	✓	✓	✗	✗
Weather / park context	✓	✓	✓	✗	✗
Remaining-runs fatigue term	✓	✗	▪	▪	✗
Forecast encompassing	✓	✓	✗	✗	✗

■ cleared this gate
 ■ failed here
 ■ not reached

Figure 2. Sequential elimination of candidate public-information hypotheses across five escalating gates. Green = cleared this gate; red = failed here; grey = not reached.

Figure 3 quantifies the attrition as a funnel: of ten hypotheses tested, nine produce a detectable in-sample association with runs, three survive out-of-sample validation, and zero add information beyond the market or clear a profitability threshold. The collapse between "survives out-of-sample" (three) and "adds information beyond the market" (zero) is the paper's pivot: predicting runs is common; predicting the market's error is the empirical boundary. The funnel counts are derived directly from the Figure 2 matrix, so the two figures cannot diverge.

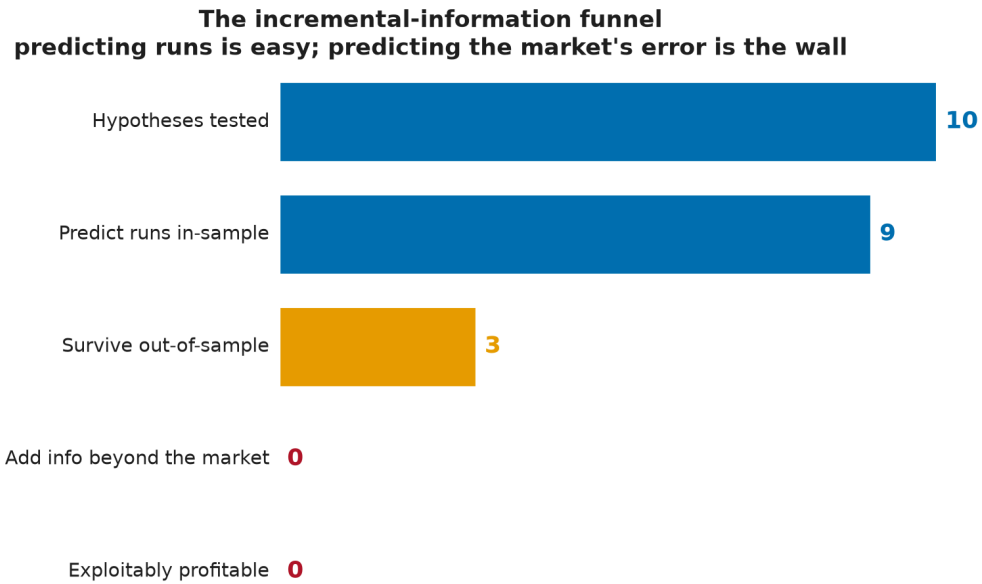


Figure 3. The incremental-information funnel. Counts derive from the Figure 2 matrix.

Figure 5 explains why one hypothesis appeared to survive before it was eliminated, and in doing so illustrates a general statistical principle. A model that adds a starter's velocity decline measured from the first to the third time through the order raises the out-of-sample AUC for "team scores above 4.5" from a tier-only baseline of 0.420 to 0.610 — an apparently large gain. But that velocity-drop variable is *post-treatment*: it is defined only for starters who survived long enough to face the order a third time, precisely the starters already being hit. Re-measuring velocity decline within a pre-treatment early window (first twenty pitches versus the next twenty) collapses the AUC to 0.524, whose Hanley–McNeil confidence interval straddles the 0.500 coin-flip line. The apparent 0.61 signal was survivorship, not fatigue — a conditioning artifact that the escalating protocol is designed to catch.

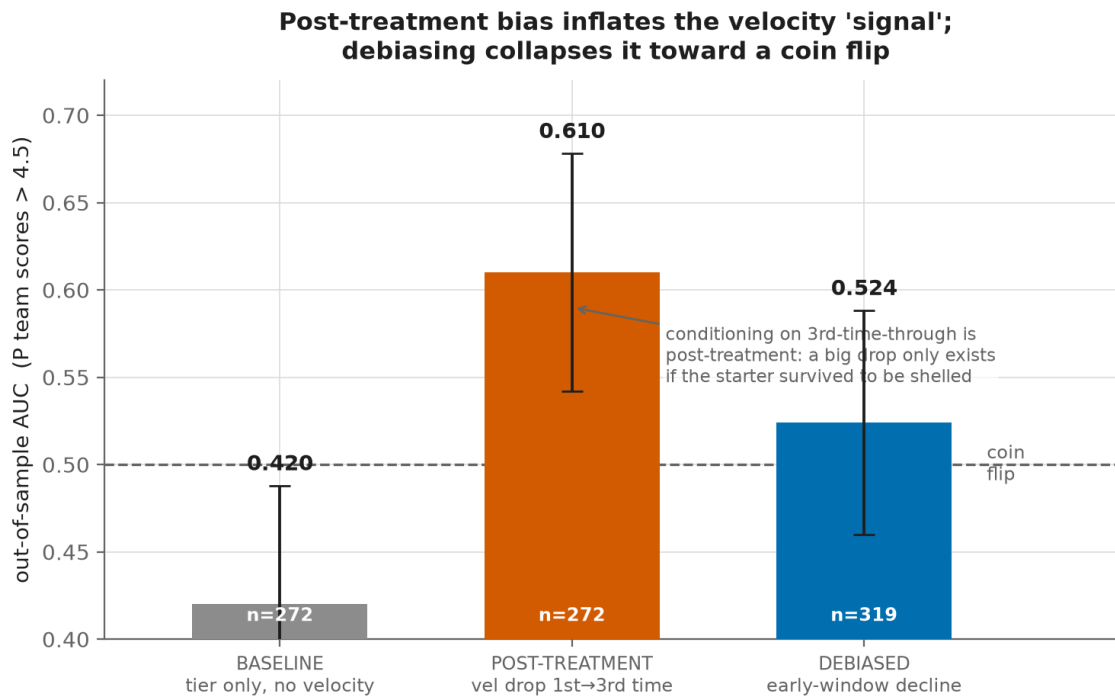


Figure 5. Post-treatment bias in the velocity signal. Out-of-sample AUC with Hanley–McNeil 95% intervals; the debaised estimate straddles the coin-flip line.

Figure 6 turns from whether the market prices information to whether it prices it by the right *amount*. For each of 6,414 in-game events we compute the true change in run expectancy (ΔRE , from a base-out RE24 table plus runs scored) and the converged move in the live total five minutes later ($\Delta Book$). Plotted against each other, every positive-run event type lies on a single common slope of approximately 0.74 through the origin; the response ratios for the higher-frequency events cluster in a narrow 0.63–0.84 band (walk 0.63, double 0.64, single 0.72, home run 0.81, triple 0.84) with no ordering by event magnitude. That uniformity is evidence about the *measurement pipeline* — a single Pinnacle-grade source sampled at roughly one-minute cadence acts as a low-pass filter that attenuates every shock by the same factor — rather than a per-event mispricing an edge could exploit. (Hit-by-pitch, $n = 115$, sits below the band; pitching changes move the line near zero, but RE24 cannot benchmark reliever quality, so we exclude them from the elasticity claim.)

**Every event type lies on one common slope
uniform attenuation (a measurement low-pass filter), not a per-event edge**

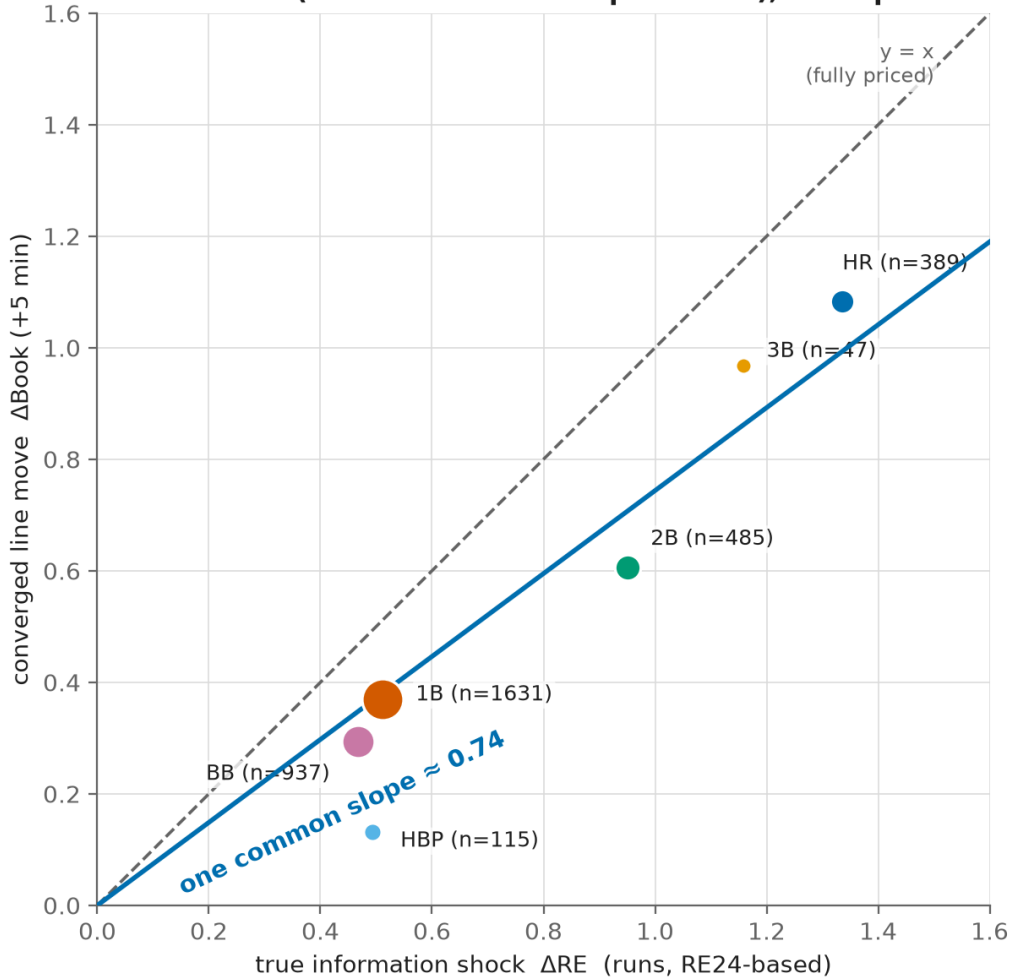


Figure 6. The market transfer function. Every positive-run event type lies on one common slope ≈ 0.74 ; marker area is proportional to event count.

Figure 7 closes the loop on the model side of the comparison. The left panel bins the 2,505 half-inning snapshots by market-implied remaining runs and plots the mean realized remaining runs in each bin: the points track the diagonal, so the market forecast is approximately calibrated within this sample (the underlying leave-one-game-out remaining-runs model, fit on 2,859 snapshots, reaches $R^2 = 0.226$, and adding fatigue terms changes its mean absolute error by -0.001 runs). The right panel shows the distribution of the book's forecast error: symmetric, centered near zero (mean $+0.49$ runs), and — as Figure 4 establishes — not predictable from any feature we measure. Together with Figure 4, this defines the empirical boundary of public-information prediction against a sharp live market.

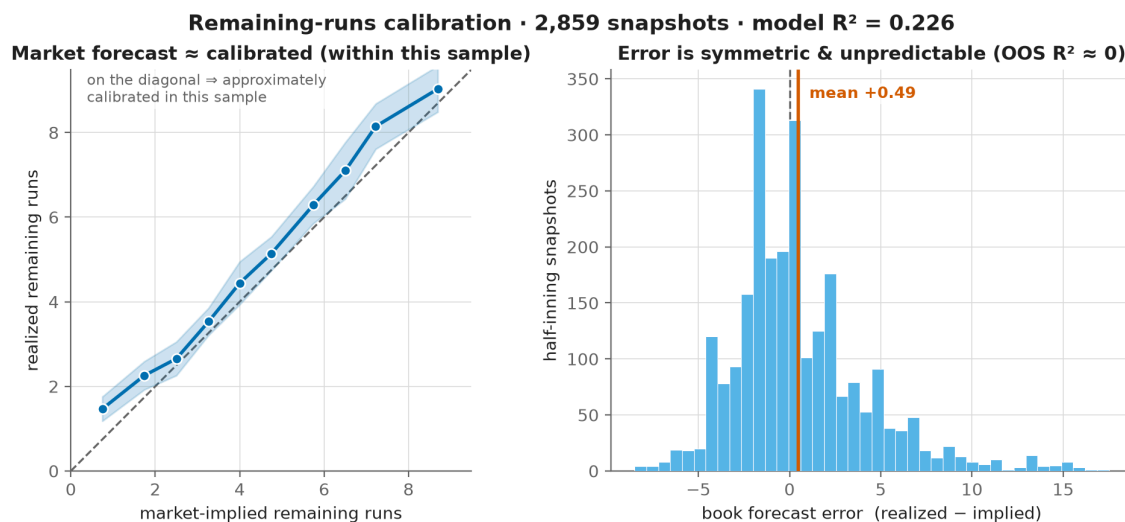


Figure 7. Market calibration. *Left:* binned market-implied versus realized remaining runs. *Right:* the distribution of the market's forecast error.

Summary of results

Across every stage of the analysis, variables that predicted runs failed to provide incremental predictive information once conditioned on the live market. The remaining sections interpret this boundary, examine its methodological implications, and discuss what classes of questions remain open.

5. Discussion

The question this study set out to answer was whether publicly observable baseball state variables contain incremental predictive information about remaining runs beyond the forecast embedded in a sharp live betting market. The Results answer it: within the limits of our data, they do not. What follows interprets that boundary — what it means, why it arises, why it matters beyond baseball, and where the evidence genuinely runs out.

5.1 What the boundary actually means

The central question of this study was not whether baseball variables predict runs. They do. Pitch count, inning, pitcher quality, weather, park effects, and velocity all contain information about future scoring, and our own feature-only forecast (out-of-sample $R^2 = 0.279$) confirms it. The question was narrower and economically more meaningful: *after* conditioning on the forecast already embedded in a sharp live betting market, do those publicly observable variables provide *additional* predictive information? Within the limits of our data, the answer is no.

That distinction — introduced in Section 1 as the paper's conceptual center — disarms the most natural objection in advance. A reader may protest that weather obviously matters, or that a tiring pitcher obviously concedes more runs. Both are true. The point of forecast encompassing is not to deny that these variables carry signal; it is to ask whether their signal is *already priced*. When the market's forecast error — the difference between what actually

happened and what the line implied — cannot be predicted from any of these variables out-of-sample ($R^2 = -0.037$), the most parsimonious reading is that the information is already incorporated into the market forecast. The variables are informative about runs and redundant with the price. Prediction survives; increment does not.

Why does this happen? Not because baseball theory is wrong, but because the forecast embedded in a sharp live market already reflects these variables. Such a forecast is produced by participants with strong incentives to price observable state correctly and quickly, and the transfer-function evidence is consistent with that: the line moves in the right direction and by a stable fraction of the true change in run expectancy after every event type, with no event class systematically under- or over-priced. A forecast that adjusts proportionately to information shocks is precisely the kind whose residual should carry no recoverable public-information signal — which is what we observe.

5.2 Prediction is not profit

Here the paper stops being about baseball. **Prediction and profit are distinct statistical problems**, and conflating them is among the most common errors in applied betting research.

Much of that literature implicitly assumes a single arrow: better prediction leads to better betting. Our results break that arrow into three distinct links that must each hold independently. A variable may *predict an outcome* — velocity decline is associated with more runs. It may nonetheless carry *no incremental information* once the market forecast is conditioned upon — because the price already reflects it. And even a variable that did carry incremental information would not automatically be *profitable*, because profitability additionally requires that the edge exceed transaction costs, survive the vig, and persist after the act of betting moves the line. Prediction, increment, and profit are three questions, not one.

Forecast encompassing is valuable precisely because it isolates the middle link — the one betting papers most often skip. A naïve backtest measures something closer to prediction; a profit-and-loss simulation measures something closer to the third link and is easily flattered by overfitting and by using stale lines. Encompassing asks the hard question directly: *does this variable improve on what the price already reflects?* In our data, for every variable we measured, the answer is no — and because the test speaks to increment rather than to a particular staking scheme, that conclusion does not depend on how one would have bet.

5.3 The efficient frontier of public information

We give this boundary a name. We use the term **efficient frontier of public information** to describe the point at which additional publicly observable variables cease to improve prediction after conditioning on the market forecast. Inside the frontier lie the observable baseball variables — pitch count, tier, bullpen, park, weather, velocity, times-through-order — that the market encompasses. Outside it lie the dimensions our data cannot reach: the timing of price formation, disagreement across books, and the evolution of the full implied distribution rather than its mean.

Every hypothesis in this study was, in effect, an attempt to move beyond that frontier using public state variables. None succeeded. That uniformity is not a series of independent disappointments; it is a single, coherent mapping of where the frontier sits for one sport, one month, and one class of information. The contribution is conceptual as

much as empirical: the frontier reframes "we failed to find an edge" as "we located the line beyond which public information stops helping," and it tells the next researcher where *not* to dig.

5.4 The methodological contribution

Although motivated by baseball, the methodology is not baseball-specific. The contribution is an escalating validation protocol designed to distinguish variables that predict outcomes from variables that contain incremental information beyond an existing forecast. Each rung strips away one more class of illusion — overfitting, then selection, then confounding, then redundancy-with-the-market — so that a hypothesis surviving to the top has been tested against progressively harder alternatives rather than a single easy one.

The reason a protocol like this matters is a matter of where the burden of proof sits. Sports betting research frequently terminates after the discovery of an in-sample signal. The present study instead treated each positive result as a hypothesis requiring progressively stronger attempts at falsification. The protocol therefore shifts the burden of proof from demonstrating *prediction* to demonstrating *incremental information beyond an existing market forecast* — a higher and more economically meaningful standard, and one that a single backtest can never meet. That shift is a philosophy of evidence, not merely a workflow.

Signal → Robustness → Out-of-sample → Debiasing → Conditional testing → Forecast encompassing → Transfer function

The ladder transfers unchanged to any market with a sharp public forecast and observable state: NBA totals, NFL spreads, soccer in-play, tennis, racing. A researcher there can adopt the same sequence, report at which rung each candidate variable is eliminated, and compare results across domains. To facilitate reproducibility, we release the datasets, evaluation protocol, and reference implementations accompanying this study as the initial release of the **Third Turn Benchmark**, so that future hypotheses can be evaluated against the same reference rather than re-derived from scratch. Its guiding principle is the same sentence that organizes the protocol (§3.3), and over time a shared benchmark and a citable protocol may prove more durable than any single finding, because they let a field accumulate falsifications instead of scattered one-off backtests.

6. Limitations

We state the conditions under which our conclusion holds, without interpretation. **Scope.** The study covers 163 games over a single month (June 2026) of one sport; the boundary is characterized precisely, but only under those conditions, and we make no claim of seasonal or cross-sport generality. **Odds source.** Historical line trajectories derive from a single Pinnacle-grade feed sampled at roughly one-minute intervals; we therefore cannot separate genuine price-formation latency from feed cadence, and the uniform sub-one response ratio in the transfer function is consistent with either. **Single-book benchmark.** All encompassing tests are conducted against one sharp book; we cannot test cross-book agreement or leadership. **Market coverage.** Retail live team totals were not exposed by our feeds and are untested, as are first-five-inning totals. **Ground truth.** The remaining-runs model and RE24

transfer benchmark use published static run values, not park- or season-specific re-estimation; the pitching-change response is reported but excluded from the elasticity claim because RE24 cannot price reliever quality. **Estimation.** Out-of-sample figures are leave-one-game-out; effective sample sizes for the rarer event types (e.g. triples, $n = 47$) are small, and the corresponding response ratios should be read with that in mind. None of these conditions is load-bearing for the central result — the book's forecast error is unpredictable from every feature we measure — but each bounds how far it may be generalized.

7. Remaining Questions

Distinct from the limitations above, this section lists what the evidence *genuinely does not answer* — the questions that remain open not because our experiment was narrow, but because they require data of a kind the historical record cannot provide. Does information propagate across books with a measurable lag, and is a laggard ever tradable? Does the market update the *shape* of the implied run distribution — its variance, skew, and tail — as accurately as it updates the mean, or is higher-moment miscalibration the place a residual edge could still hide? What is the information half-life of a given shock: how long does the line take to absorb a home run versus a pitching change versus an injury? Does the boundary we map for full-game totals move when the market isolates the starters, as first-five-inning totals do? Each of these is a live-data question — none is answerable from one-minute historical snapshots of a single book — and each is the natural subject of the market-microstructure study that our forward-collected, timestamped streams are being built to support.

8. Conclusion — what we learned

This project began with a search for an exploitable feature of baseball. It ended by identifying the empirical boundary at which publicly observable baseball information ceases to provide incremental predictive value against a sharp live market. That boundary is itself a result. It redirects future work away from discovering additional baseball variables and toward understanding how information propagates through live betting markets. The contribution of this study is therefore not a successful betting strategy, but a reproducible framework for determining when one does — and does not — exist.

Appendix Table A1 — the full hypothesis battery

Referenced once from §4; Figure 2 is the main-text representation.

Hypothesis	Motivation	Test	Outcome	Mode of elimination
Times-through-order	Familiarity/fatigue penalty on 3rd time through	Binary gate → gradient, LOGO	Encompassed	Decay is continuous, not a discontinuity; out-of-sample fires below breakeven; market prices it

Hypothesis	Motivation	Test	Outcome	Mode of elimination
Velocity decline	Fatigue shows as lost velocity	Debiased early-window vs post-treatment	Artifact	Defined only for starters who remained in long enough to be measured (selection); clean signal AUC ≈ 0.52
Bullpen fatigue	Fatigued bullpen $\rightarrow$ higher scoring after a starter collapse	Isolated to the bullpen's own innings	No effect	Fatigued bullpens concede the same or fewer runs; no multiplier
Drop reversion (Over)	Over-dropped line reverts up	Threshold sweep, all games	Not out-of-sample	Reversion is right-skewed (win-big/lose-small); median below the line
Drop reversion (Under)	Line stays low after a slow start	Banded + robustness gates	Not robust	Effective band moves with the snapshot inning; concentrated in the recent sample
Alternate-line skew	Buy the fat upper tail at longer odds	Empirical win% vs efficient-implied	Priced	Empirical $<$ implied at every half-run increment above the main line; tail priced fatter than realized
Early-run anchoring	Live total under-reacts to a 1st-inning scoring burst	Post-1st Over, cause split	Priced	49/50 bursts hit-driven (no fluky-runs population); market prices the change
Weather / park	Books under-price hitter-friendly context	Conditional split	Priced	Hitter-friendly Overs hit <i>less</i> (46% $<$ 50%); market over-adjusts for context
Remaining-runs fatigue	Fatigue adds to a state model	Incremental MAE, LOGO	No increment	Game state already contains the information; $\Delta\text{MAE} -0.001$
Forecast encompassing	Does <i>anything</i> beat the market?	$Y \sim B + X$; $(Y - B) \sim X$; per-feature E+	Encompassed	Book error not predictable from any feature out-of-sample ($R^2 \approx 0$)

Data and code availability

Cleaned datasets, feature schema, evaluation protocol, reference implementations, and frozen result artifacts are released as the initial release of the Third Turn Benchmark, archived under a persistent DOI alongside the code repository and the research log. (*DOI + benchmark packaging pending.*)

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